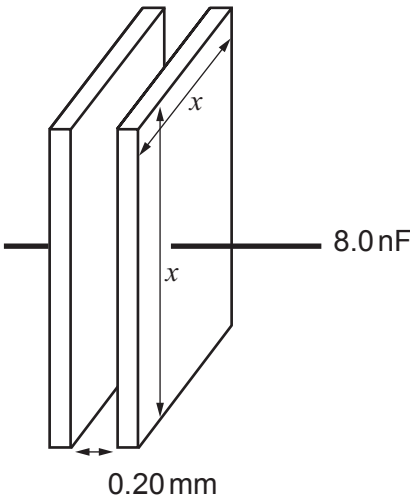


SECTION A

Answer all questions.

1. (a) (i) An 8.0 nF capacitor has square plates separated by 0.20 mm of air. Calculate the length (x) of the sides of the plates. [3]



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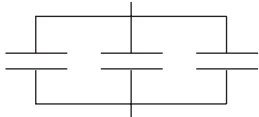


(ii) Which of the following combinations of 8.0 nF capacitors has a capacitance of 12.0 nF? Justify your answer. [4]

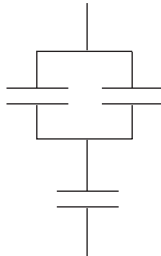
All individual capacitors shown are 8.0 nF.



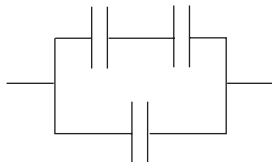
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(iii) State how the capacitance of the capacitor can be increased without changing its dimensions. [1]

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(iv) Calculate the charge on the plates of the 8.0 nF capacitor when it stores an energy of 0.62 μJ. [3]

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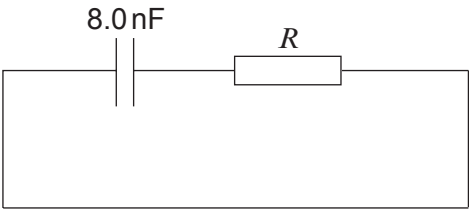
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- (b) (i) A charged 8.0 nF capacitor is discharged through a large resistor. Calculate the resistance of the resistor if the time constant is 15.5 ms . [2]



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- (ii) Calculate the fraction of the initial charge remaining on the capacitor after 31.0 ms . [3]

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